Continuous Distillation

Final Lab Report

Unit Operations Lab 5

April 06, 2026

Group 3

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1. Executive Summary

Using the Armfield UOP3CC Continuous Distillation Column for a 10wt% ethanol reboiler solution and 20 wt% ethanol feed solution, the main objective of this lab is to study the effect of feed stage on the column efficiency via the McCabe-Thiele method under steady-state conditions and by varying the feed position. A summary of the number of stages calculated from the McCabe-Thiele method and column efficiency for each operating condition is provided below.

Table 1: Summary of Calculated Values

Operating Conditions	Number of Stages	Column Efficiency, %
Steady-State Distillation	6	75
Varying Feed Position	3	37.5

Overall, McCabe-Thiele analysis showed that steady-state distillation has a higher number of stages, and therefore has a higher column efficiency compared to varying feed position, which is an expected behavior.

2. Introduction

The main goals of this lab are to plot the temperature of all eight thermocouples as a function of time, as well as to investigate the effects of the feed stage on the column efficiency using the McCabe-Thiele analysis. Continuous distillation experiments are crucial for engineers to understand, as it's commonly used in the pharmaceutical industry for purification and in laboratory settings [1]. For the experiment, an Armfield UOP3CC Column was used as the primary unit for the continuous distillation of the ethanol-water mixture, the control system was used to control the column/reboiler specifications, the condenser was used to condense the overhead vapor, reboiler was used to contain and heat the solution, a peristaltic feed pump was used to continuously feed solution into the system, feed tank was used to hold the feed solution, an accumulator was used to collect the distillate, a top product receiver was used to collect the overhead product, a heat exchanger was used to warm up the feed to the column, a drain valve was used to drain the reboiler, water flow meter was used to monitor the flow rate of water to condenser, and a pressure gauge was used to monitor the pressure of the column. To prepare for the experiment, the feed pump was first calibrated, which was performed by the power of the pump motor from 0-10 at an increment of 1. At each increment, water was allowed to flow into a graduated cylinder for 30 seconds, and the flow rate was calculated. Upon gathering enough data, Excel's "LINESST" function was used, allowing the slope of the curve to be calculated. After the pump was calibrated and ensuring the drain valve was closed, a 10L solution containing 10wt% ethanol and 90wt% water was made and transferred to the reboiler. A 3L solution of 20 wt% ethanol and 80 wt% water was then made and transferred to the feed tank. The feed pump was installed, and its head was then connected to the middle of the column. The control panel was then turned on, to which the condenser's cooling water flow rate was set to about 3L/min and valve V5 was opened. Experiment B then began by setting the power to 1.5kW and the system was allowed to reach steady-state, which is shown by small changes in temperature of thermocouples T1-T8. After reaching steady-state, the power was then set to 0.75 kW, reflux ratio was set to 5:1, valve V1 was opened, and the system was allowed to reach steady-state once more. After approximately 30 minutes, the feed pump was turned on, and approximately 400mL of feed solution was allowed to flow into the column. Then, overhead and bottom samples were simultaneously taken via valves V3 and V2, respectively. The overhead and bottom samples were taken two more times, with a 5-minute increment each time. The temperature from thermocouples T1-T9 was also recorded for each trial. Ethanol content in the samples was then analyzed using a clean and calibrated Snap41 densitometer. The pump head was then removed from middle of the column to the top of the column, and Experiment C then followed an identical procedure described above.

3. Theory

Continuous distillation separates a binary mixture by repeatedly contacting the rising vapor with descending liquid inside the column [3][5]. In this ethanol-water setup, the feed enters near the middle, and the reboiler adds heat near the bottom. For this to generate vapor; The condenser at the top turns overhead vapor back into liquid. Parts of this condensate is returned as the reflux while the rest is collected as distillate [3][4]. The main idea is for each tray/stage to push the vapor and liquid compositions closer to the vapor-liquid equilibrium so that the top eventually becomes richer in the more volatile component (ethanol), while the bottom becomes richer in the other (water) [3][5].

The column is described with mass and energy balances [3][4]. At minimum, an overall mass balance, component balances, stage-by-stage balances, and an overall heat balance for the system are needed [3][4]. In theory, these balances show how feed composition, product flow rates, and heat input control the separation. The two biggest operating variables are the reflux ratio and feed condition [3][4][5]; A higher reflux ratio usually improves separation because more liquid is returned to contact the rising vapor, but it also increases the energy demand on the reboiler. The feed condition matters too, because a saturated liquid feed affects the internal liquid and vapor traffic differently than a partially vaporized feed would [2][3][5].

For a binary system (ethanol-water), the main design tool is the McCabe-Thiele method [3][5]. This method uses the vapor-liquid equilibrium curve plotted as vapor composition, , vs. liquid composition, , along with operating lines based on stagewise mass balances. For a continuous column with feed entering the middle, there is a rectifying operating line between the equilibrium curve and the operating lines to estimate the number of ideal stages (for the desired separation) [3][5].

That is where column efficiency comes in. the ideal McCabe-Thiele analysis assumes each tray reaches equilibrium; however real trays do not follow true to this assumption. The Murphree plate efficiency is used to measure how close an actual tray approaches ideal equilibrium behavior. If the trays are < 100% efficient, the real column needs more actual stages than the ideal calculation predicts. On a McCabe-Thiele diagram this shows up as an effective shift away from the true equilibrium curve; So, if the experimental compositions do not match the ideal prediction, one major reason is that the tray efficiency is less than unity [3][4][5].

Feed position also changes the behavior of the column. In this apparatus, the feed can enter at different locations, which changes how much of the column acts mainly as a rectifying/stripping section. Moving the feed above or below the normal midpoint changes the internal composition profile and can weaken the separation if the feed is not introduced near its best stage [3][5].

Table 2: Variables and their Descriptions

Variable	Units	Description
F	mol/s or kmol/h	Feed flow rate entering the column
	dimensionless	Ethanol mole fraction in the feed
D	mol/s or kmol/h	Distillate flow rate collected at the top
	dimensionless	Ethanol mole fraction in the distillate
B	mol/s or kmol/h	Bottoms flow rate removed from the bottom of the column or reboiler
	dimensionless	Ethanol mole fraction in the bottoms product
L	mol/s or kmol/h	Internal liquid flow rate in the column
V	mol/s or kmol/h	Internal vapor flow rate in the column
	mol/s or kmol/h	Reflux flow rate returned from the condenser to the column
R	dimensionless	Reflux ratio, usually defined as $R = L0/D$
	kW or kJ/h	Reboiler heat input supplied at the bottom
	kW or kJ/h	Condenser heat removed at the top

x	dimensionless	Liquid-phase ethanol mole fraction on a tray or stage
y	dimensionless	Vapor-phase ethanol mole fraction on a tray or stage
	dimensionless	Liquid composition leaving stage *n*
	dimensionless	Vapor composition leaving stage *n*
n	stage number	Index used to identify a tray or equilibrium stage
N	number of stages	Total number of ideal stages required by the McCabe-Thiele method
	number of trays or stages	Actual number of trays needed in the real column
	stage number	Feed tray location
q	dimensionless	Feed condition parameter showing how much of the feed is liquid versus vapor
or	dimensionless or %	Murphree plate efficiency showing how closely a tray approaches equilibrium
α	dimensionless	Relative volatility of ethanol to water
T	°C or K	Temperature of a stage, feed, condenser, or reboiler
P	kPa, bar, or atm	Operating pressure of the column
	kJ/mol or kJ/kmol	Feed enthalpy
	kJ/mol or kJ/kmol	Distillate enthalpy
	kJ/mol or kJ/kmol	Bottoms enthalpy
	kJ/mol or kJ/kmol	Enthalpy of liquid stream in the column
	kJ/mol or kJ/kmol	Enthalpy of vapor stream in the column

4. Results & Data Analysis

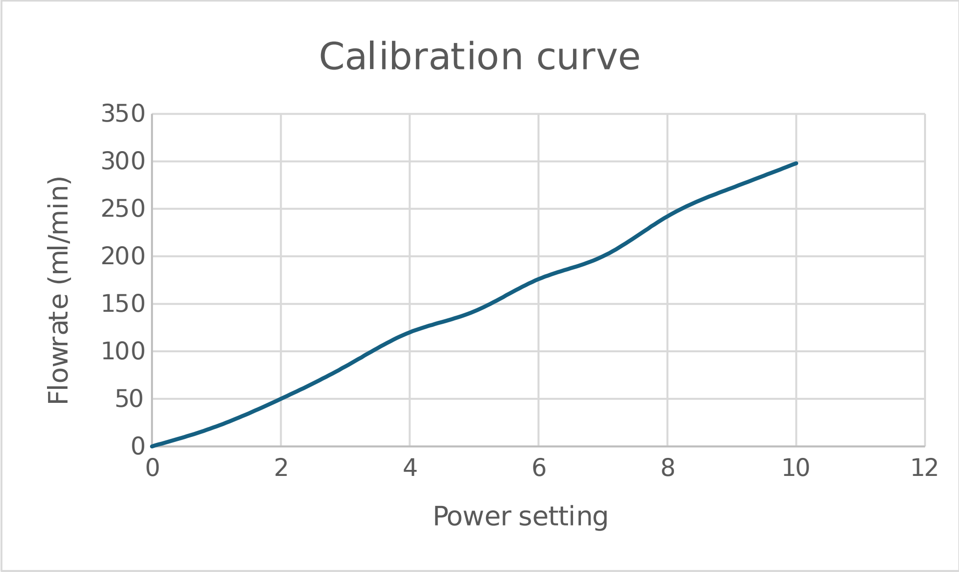

Figure 1: Feed Pump Calibration Curve

The feed pump calibration curve shows how the flow entering the column corresponds to the pump setting on the controller. At max speed on the controller (10), a flow rate of 300 ml/min was achieved, which progressed for each integer speed setting to achieve the above calibration curve for the experiment.

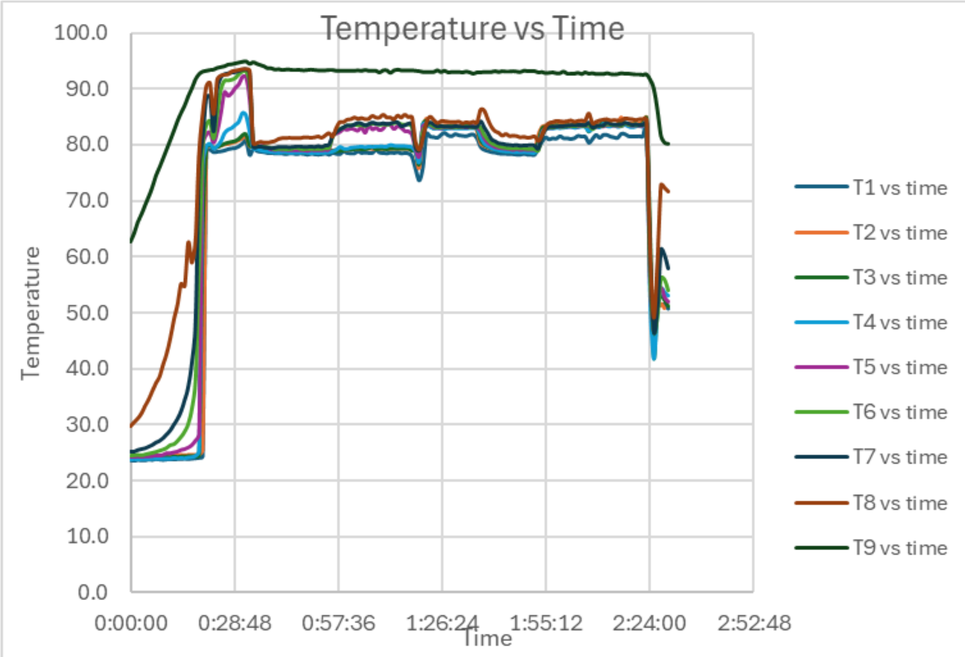

Figure 2: Temperature as a Function of Time

The overall temperature record from the experiment is shown above. As seen in Figure 2, after the system achieved steady state at the ~30-minute mark, the first experiment was conducted, and after switching the feed location and waiting for the system to stabilize at the 1:55 mark, the second experiment was conducted.

To verify the column had reached steady state, three samples were taken from both the distillate and reboiler 5 minutes apart to ensure steady state. For experiment B, the average ethanol concentration was 92.16 wt% for the top product and 8.58 wt% for the bottom product, which when converted to mol% are 78.5mol% and 2.83%, respectively. For experiment C, the results were 83.87 wt% for the top product and 10.15 wt% for the bottom, corresponding to 61.7mol% and 3.37mol%. These results show that the middle feed configuration of the column produced a better separation.

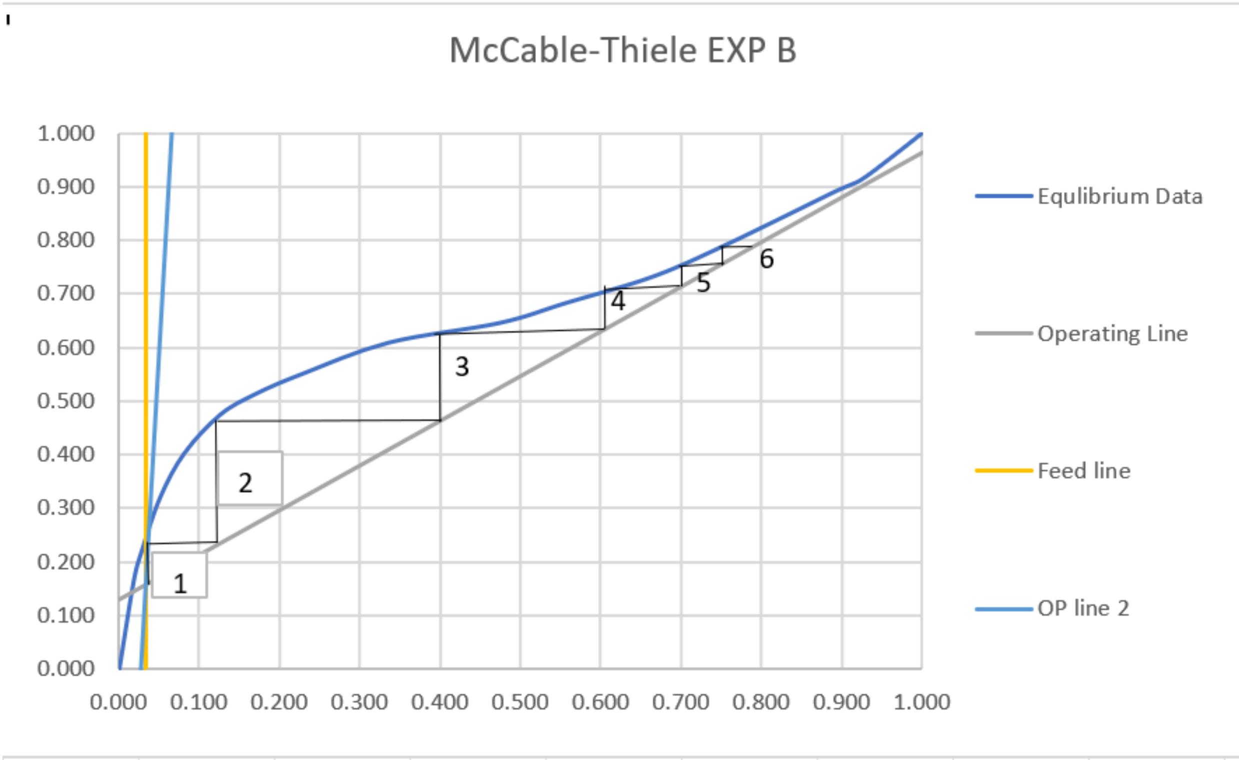

Figure 3: McCabe-Thiele Diagram of Experiment B

The McCabe-Thiele diagram for experiment B, middle feed location, is used to compare the ideal behavior of the column with the experimental separation performed. The equilibrium data was taken from ASPEN PLUS V14.0 [6]. The equilibrium curve shows the compositions that exist on each stage when the vapor and liquid reach equilibrium, while the operating lines show the mass balance in the system at a given reflux ratio and feed. The step-off procedure between the operating lines and equilibrium curve gives six ideal stages for this separation, in contrast with the eight physical stages present. In reality, the actual trays present in the column do not reach equilibrium, so the real column performance is less efficient than the ideal.

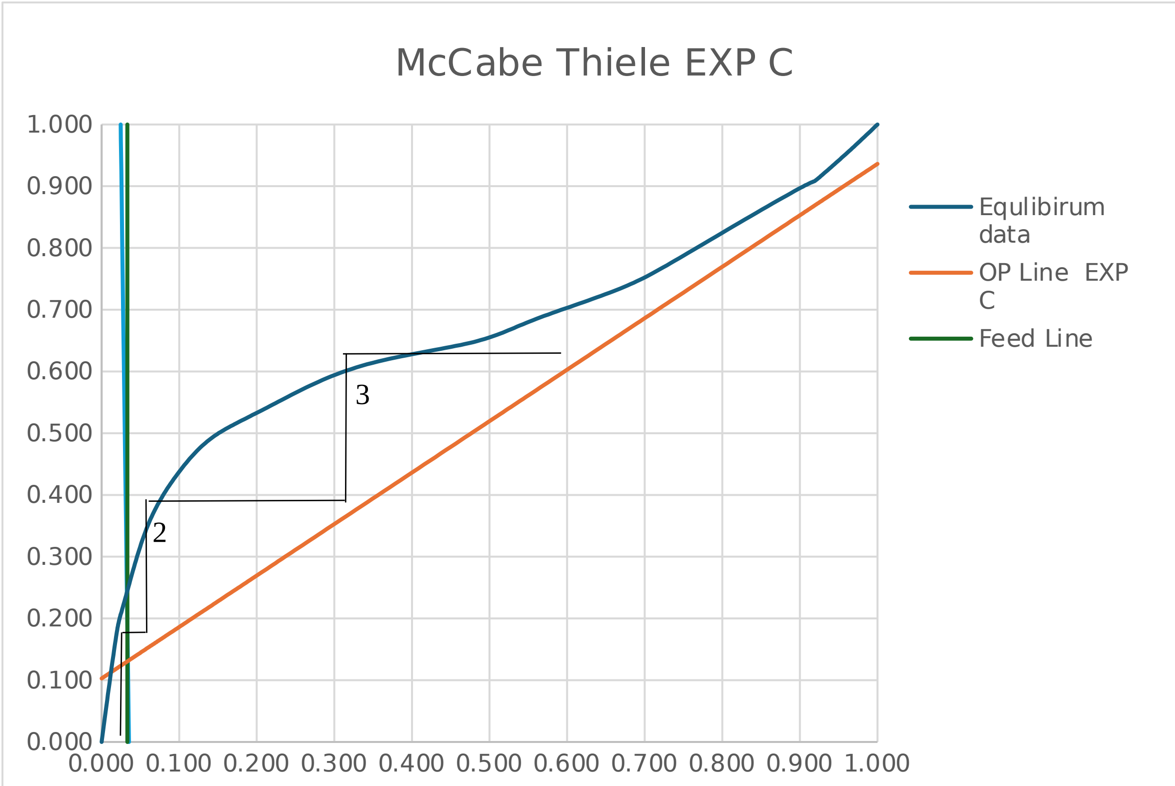

Figure 4: McCabe-Thiele Diagram of Experiment C

For experiment C, the feed location was moved to above the top tray. This changed the operation of the column. This change decreased the effectiveness of the separation compared to experiment B. The McCabe-Thiele diagram shows three ideal stages present based on the steps between the equilibrium and operating curves. This data suggests that the new feed position made the column behave more like a stripping section and reduced how well ethanol and water could be separated.

The overall column efficiency is calculated by comparing the number of ideal stages from the McCabe-Thiele analysis to the eight actual trays present in the column. For experiment B six ideal stages were obtained at an efficiency of 75% and for experiment C three ideal stages were obtained at an efficiency of 37.5%. These results show that the column performed much better when the feed was introduced at the midpoint instead of above the top tray. Since the analysis assumes vapor-liquid equilibrium being present on each stage, the lower efficiency in experiment C shows that the change in the feed location corresponded to a drop in efficiency.

An overall energy balance was performed at steady state conditions using the reboiler and condenser duty. The following assumption is outlined in the lab manual for this analysis;

the energy carried by the feed is balanced by the energy leaving through the top and bottom, allowing the energy balance to the reboiler and condenser. The reboiler duty at steady state is 0.75 kW and the condenser duty is 0.649 kW calculated from the inlet and outlet of the condenser cooling water, 13.4 C and 16.5 C. The delta between the condenser duty is 0.101 kW with a heat loss of 13.5% suggesting that the column cannot be assumed to be adiabatic and the heat losses to the surroundings contributed to the differences between the McCabe-Thiele analysis and the column.

5. Discussion

The results of this experiment are generally consistent with distillation theory, particularly regarding the effect of feed location on separation performance. In experiment B, where the feed was introduced at the midpoint, the column achieved a top composition of 78.4 mol% ethanol and a bottom composition of 2.83 mol%. In experiment C, with the feed introduced above the top tray, the top composition decreased to 61.7 mol% while the bottom composition increased slightly to 3.37 mol%. This trend agrees with theory, as optimal feed placement occurs near the stage where the feed composition matches the internal column composition, allowing both rectifying and stripping sections to operate effectively.

The Mcabe-Thiele analysis supports these observations. Experiment B resulted in six theoretical stages and an overall efficiency of 75%, while experiment C only had three theoretical stages with an efficiency of 37.5%. These values fall within expected ranges for tray efficiencies in small scale columns. The reduced efficiency in experiment shows the negative impact of improper feed placement. This highlights how feed location directly influences the effective number of stages contributing separation

Despite agreement with theoretical trends, some errors are present in the results. The small change in bottom composition between the two experiments suggests that the stripping section performance may not have been significantly affected. Additionally, fluctuations are observed in the temperature vs. time data, which may indicate that steady-state conditions may not have been fully achieved. This can introduce errors in the composition measurements and calculated efficiencies.

Overall, the results demonstrate that feed location has a significant impact on column performance, with midpoint feed providing a much better separation efficiency. This aligns with industrial distillation practices, where proper feed tray selection is important. Future work could involve testing additional feed locations, varying reflux ratio, and comparing results with simulation models to better know any deviation from ideal operation.

6. References

[1] Aishengke. (2023, January 3). Advantages and applications of continuous distillation unit. https://www.aishengk.com/news_xq/32.html
[2] J. Kunesh, H. Kister, and et al., “Distillation: Still towering over other options,” *Chemical Engineering Progress*, vol. 89, p. 43, 1995.
[3] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005.
[4] R. H. Perry and D. W. Green, Eds., *Perry’s Chemical Engineer’s Handbook*, 7th ed. McGraw-Hill, 1997.
[5] P. C. Wankat, *Equilibrium-Staged Separations*. Prentice Hall, 1988.
[6] Aspen Plus V14.0

7. Appendix I: Data Tabulations/Graphs

NOTE: Raw data of temperature from thermocouples T1-T9 will be submitted along with the report

Exp B											
Time,min	Snap 41 top, %	Snap 41 bot, %	T1,C	T2,C	T3,C	T4,C	T5,C	T6,C	T7,C	T8,C	T9,C
0	91.95	7.68	78.7	79.3	79.3	79.7	82.7	83.9	84.2	85.1	93.3
5	92.28	9.03	78.6	79.3	79.2	79.8	83.7	83.6	83.8	84.8	93.4
10	92.25	9.03	78.7	79.4	79.3	79.8	82.4	83.6	83.6	85.3	93.4

Exp C											
Time,min	Snap 41 top, %	Snap 41 bot, %	T1,C	T2,C	T3,C	T4,C	T5,C	T6,C	T7,C	T8,C	T9,C
0	84.3	9.74	81.4	83.6	83.4	83.2	83.3	83.3	83.5	84.3	92.7
5	83.93	10.43	81.4	83.7	83.6	83.3	83.3	83.6	83.7	84.3	92.8
10	83.38	10.28	81.7	83.9	83.6	83.4	83.3	83.6	83.6	84.5	92.7

Figure 5: Raw Data from Snap41 Densitometer

Power	Water amount, mL	Time, s	Flow rate, ml/s	Flow rate, ml/min			
0	0	0	0	0			
1	10.5	30	0.35	21			
2	25	30	0.833333	50			
3	42	30	1.4	84			
4	60	30	2	120			
5	71	30	2.366667	142			
6	88	30	2.933333	176		30.52727	-6.72727
7	100	30	3.333333	200		34.48455	
8	121	30	4.033333	242			
9	136	30	4.533333	272		1.35	
10	149	30	4.966667	298			

Figure 6: Raw Data for Calibration Curve

8. Appendix II: Error Analysis

Both quantitative and qualitative sources of error were present in this experiment. Quantitatively, the interpretation of the McCabe-Thiele method. The feed flow rate is also estimated from a calibration curve, which can be used as the operating line. The graphical stepping procedure can introduce some uncertainty for the number of theoretical stages. This can result in a difference in tray efficiency. While these uncertainties affect numerical values, they do not change the overall trend observed in both experiments.

Qualitatively, the most significant source of error can come from heat loss from the column and some uncertainty in the reflux ratio. Temperature fluctuations suggest the system may not have been perfectly stable while sampling, which could affect measured compositions. Heat loss causes deviations from the ideal behavior that we assumed. The reflux ratio was also controlled by timed switching rather than direct flow measurement, which can introduce some uncertainty.

Overall, these errors primarily impact the precision of calculated efficiencies rather than the validity of the conclusion. The results clearly show that the midpoint feed location improves separation performance compared to feeding above the top tray. Future improvements could include direct measurement of flow rates, better insulation, longer stabilization times, and more samples to reduce uncertainty and improve accuracy.

9. Appendix III: Sample Calculations

Experiment B values will be used for the sample calculations

Average top and bottom Ethanol %:

Feed Composition:

Reflux Ratio:

Operating Line:

Repeat this equation for the remaining amount of Liquid mole fraction ethanol to get the entire operation line.

Tray Efficiency:

Energy Balance:

Given the following:

Plugging into the equation for condenser removed

Using the reboiler duty the heat loss from the column is:

Percent heat loss:

10. Appendix IV: Individual Team Contributions

ASSIGNMENT	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	All Team members
Pre-Lab Editing	2	All Team members
Experimental Objective (Prelab)	1	Miguel Piza
Apparatus (prelab)	1	Khoa Nguyen & Rishabh Patel
Materials & Supplies (prelab)	1	Khoa Nguyen
Procedure (prelab)	1	Alexis Guillen
Project Safety Evaluation (prelab)	1	Kyle Hankosky
Final Lab Editing	2	All Team members
Executive Summary (final lab)	2	Khoa Nguyen & Rishabh Patel
Introduction (Final lab)	1	Khoa Nguyen & Rishabh Patel
Theory (Final Lab)	5	Rishabh Patel & Khoa Nguyen
Results (final lab)	5	Kyle Hankosky & Alexis Guillen & Miguel Piza
Discussion (final lab)	2	Khoa Nguyen & Miguel Piza
References (final and/or	1	Rishabh Patel

prelab)		
Data Tabulation/Graphs (final)	2	Kyle Hankosky & Alexis Guillen & Miguel Piza
Error Analysis (final lab)	1	Kyle Hankosky & Alexis Guillen
Sample Calculations (final Lab)	2	Kyle Hankosky & Alexis Guillen
PowerPoint Presentation	5	All Team members
Total	39 Hours	